Resonant scattering effects of the member asteroids in Maria Family

[1]Xi-Feng Cha*, [1]Yu-Chi Cheng, [2]Hsuan-Ting Lai, [1]Wing-Huen Ip
[1]Graduate Institute of Astronomy, National Central University, 320317, Taiwan
[2]Department of Earth Sciences, National Central University, 320317, Taiwan

ABSTRACT

Near-Earth Asteroids (NEAs) come from Main Belt Asteroids (MBAs) with injection through Mars crossing. The primary of mechanism is associated with the orbital perturbation effects caused by the mean-motion resonances (MMRs) coupled with the Yarkovsky effect. Clear examples can be seen from the cut-offs of the V-shape distributions of some asteroid families. This work we detect one of MBA families Maria. In the results of either ejection or injection, we can observe the moment the members injected to NEAs through Mars crossing or ejection. Eventually, we compare two sides of MMRs (3:1 and 8:3) in Maria Family and discuss the difference between the both MMRs.

INTRODUCTION

[Origin of Near-Earth Asteroids]

Where do NEAs come from? They come from Main Belt. How do MBAs become NEAs? They are in perturbation to encounter Martian orbit primitively caused by the solar radiation whose effect calls "Yarkovsky Effect".

[Yarkovsky Effect]

As a non-gravitational effect, the generation of Yarkovsky effect is caused by solar radiation, making asteroids spin within different temperature between Sun-facing side and the opposite side[1]. The expression uses Yarkovsky drift (da/dt), the change of semi-major axis under Yarkovsky effect, which is defined as

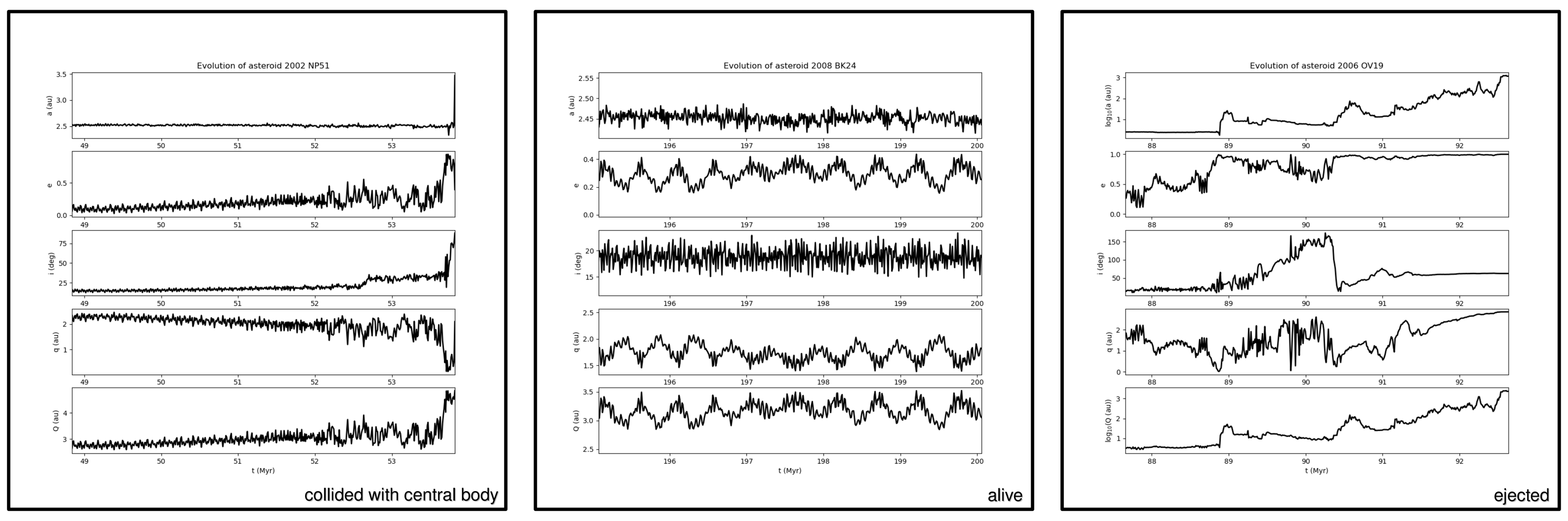
$$\frac{da}{dt}=\left(\frac{da}{dt}\right)_0\times\frac{\sqrt{a_0}(1-e_0^2)}{\sqrt{a}(1-e^2)}\times\left(\frac{D_0}{D}\right)^{\alpha}\times\frac{\rho_0}{\rho}\times\frac{1-A}{1-A_0}\times\frac{\cos\theta}{\cos\theta_0} \qquad (1)$$

which a is the semi-major axis; e is the eccentricity; D is the size of an asteroid; ρ is the density; A is the surface albedo; θ is the obliquity, and the index "0" is at zero-point at inner main belt[2].

In Equation (1) we find that when $\theta < 90°$, $da/dt > 0$ with spinning spins clockwise; when $\theta > 90°$, $da/dt < 0$ with spinning counter-clockwise. To express the Yarkovsky drift, we illustrate v-shape at an asteroid family in a-H relation for absolute magnitude H (Figure 1 for instance).

RESULTS

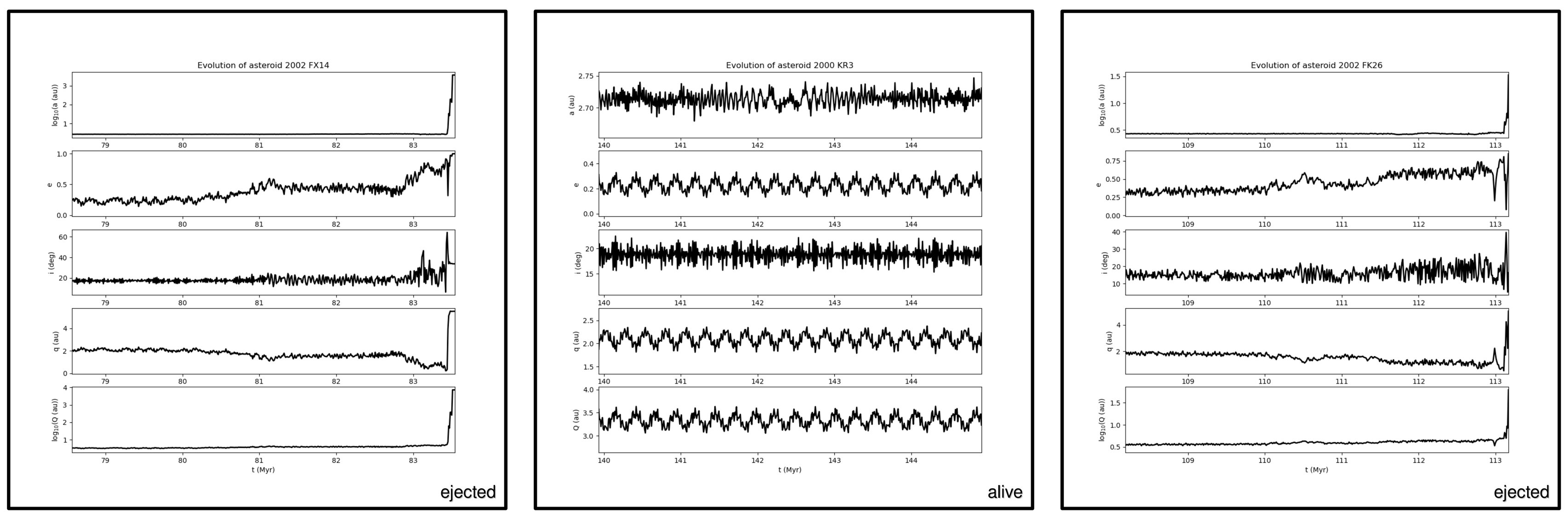
[3:1 MMR]



(a) (99849) 2002 NP$_{51}$ (b) (256746) 2008 BK$_{24}$ (c) (361175) 2006 OV$_{19}$

Figure 2: Orbital evolutions of some members in 3:1 MMR

[8:3 MMR]



(a) (126959) 2002 FX$_{14}$ (b) (60975) 2000 KR$_3$ (c) (126980) 2002 FK$_{26}$

Figure 3: Orbital evolutions of some members in 8:3 MMR

OBJECTS

[Maria Family]

Maria family is a large, old group of stony (S-type) asteroids in the inner main belt, named after asteroid 170 Maria, known for its high inclination and location near Jupiter's 3:1 resonance, potentially supplying Near-Earth Asteroids.

[Range for This Work]

This work we select around 3:1 (77 objects) and 8:3 MMR (79 objects).

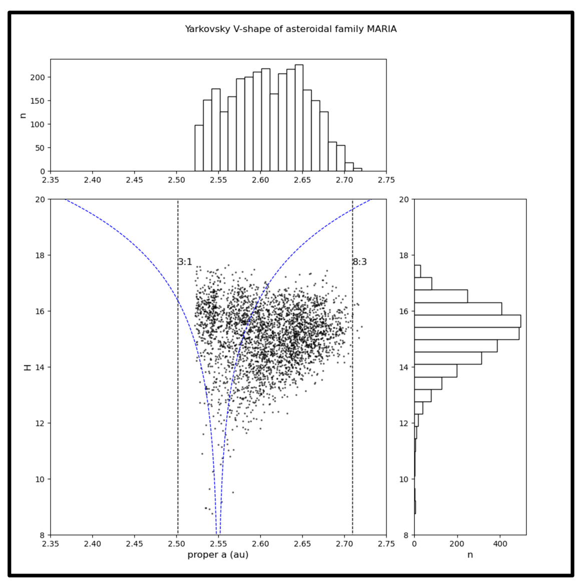


Figure 1 ►
The distribution of asteroid family Maria and its Yarkovsky V-shape curve.

METHODS

[Determine da/dt]

Assume the objects in range of this work are 1-km size. Thus the da/dt ~ -2 * 10^{-4} au/Myr for 3:1 MMR and 2 * 10^{-4} au/Myr for 8:3 MMR[2][3]. (Scan QRcode for details.)

[Initial setup]

Package	Integrator	Time step	Evolution time	$(da/dt)_{3:1}$		$(da/dt)_{8:3}$	
MERCURY	Hybrid	1 day	200 Myr	-2 * 10^{-4}	au/Myr	2 * 10^{-4}	au/Myr

DISCUSSIONS & CONCLUSIONS

[Statistical Analysis so far]

- 3:1 MMR: (#injection, #alive, #ejection) = (42, 1, 34)
- 8:3 MMR: (#injection, #alive, #ejection) = (0, 77, 2)

[Discussion and Conclusion]

Comparing with both MMRs, 3:1 MMR has earlier perturbation than 8:3 MMR's due to stronger solar radiation at 3:1. And it's clear to see how they become NEAs through Martian orbit[4]. In advance, we focus on Figure 2(c), investigating that the inclination become larger than 90°. We suspect many times encountering Jupiter. To find out the truth, we need to zoom the timeline in the future work. Also, we have to compare with other families to ensure how they evolve. Last but not least, the test is upon 1-km size, thus we will use the real size of the members to simulate the evolution.

REFERENCES

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Scan QRcode on top to see more!!!

CONTACT

Name: Xi-Feng Cha
Email: m1129004@gm.astro.ncu.edu.tw
Discord: robertcha0808

ASTROHOLIC_TREBOR

Graduate Institute of Astronomy, NCU
國立中央大學天文研究所